

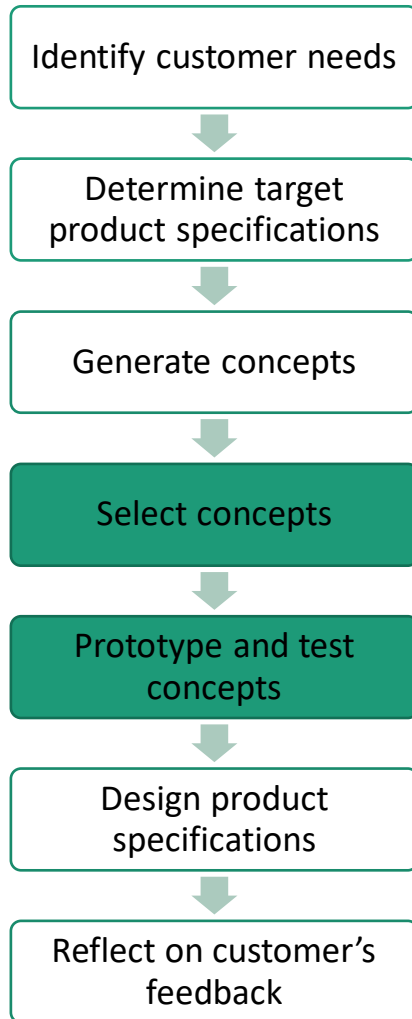
Concept Selection and Testing

Lecture 7

Outline

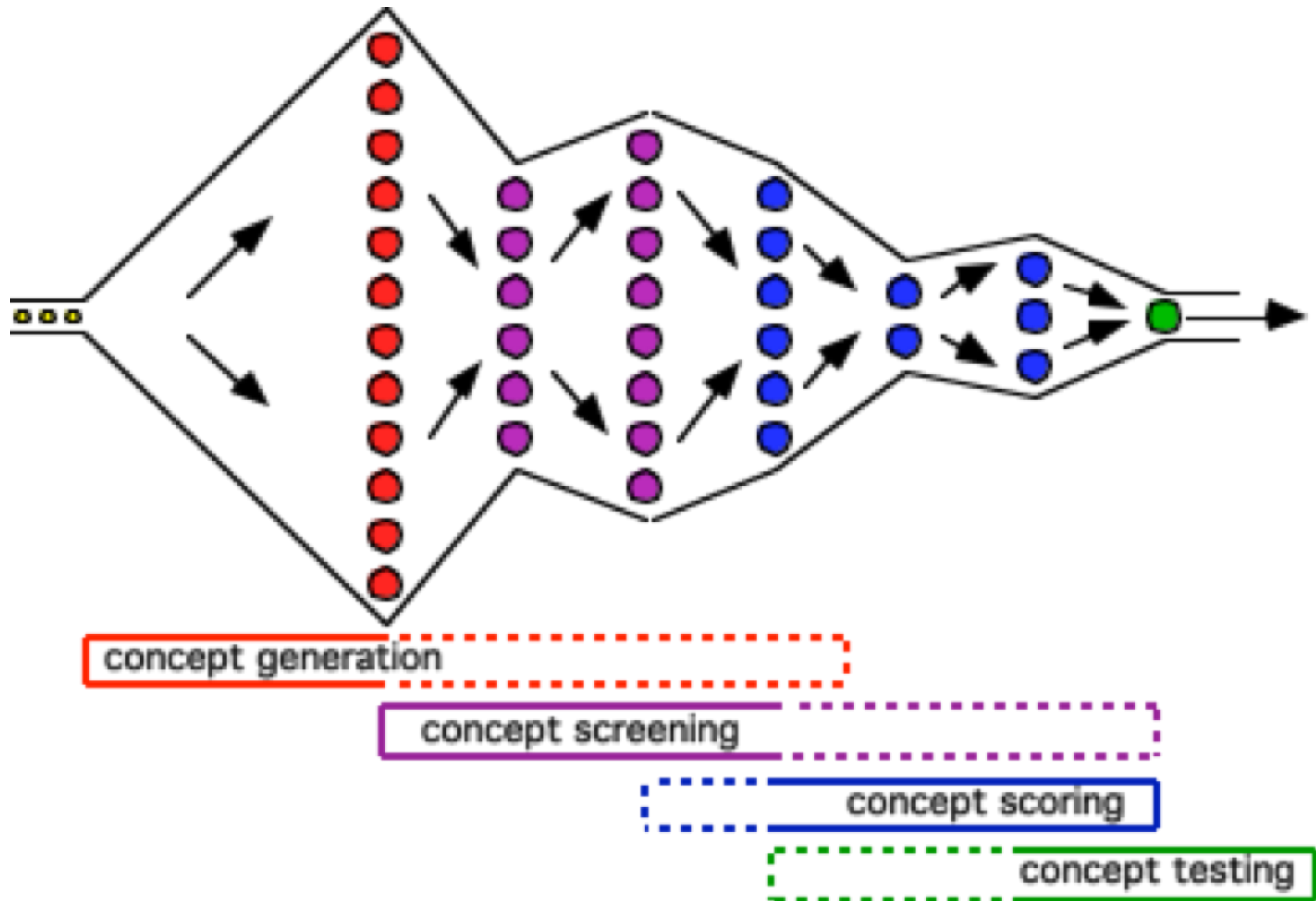
- Concept Selection Methods
- Case Study: Concept Selection
- **Classroom Activity: Decision Matrix**
- Concept Scoring with a Decision Matrix
- Concept Testing
- Engineering Idea #4: Pricing Options

Selecting and testing concepts



- Choosing the best concept, given that the designs are still quite abstract
- Making team-based decision for the best concept selection.
- Find desired attributes of otherwise weak concepts
- Defining a concept/prototype test and survey population
- Defining a survey format and measuring customer response
- Interpreting the results of the survey

Concept Development Funnel



Some Concept Selection Methods



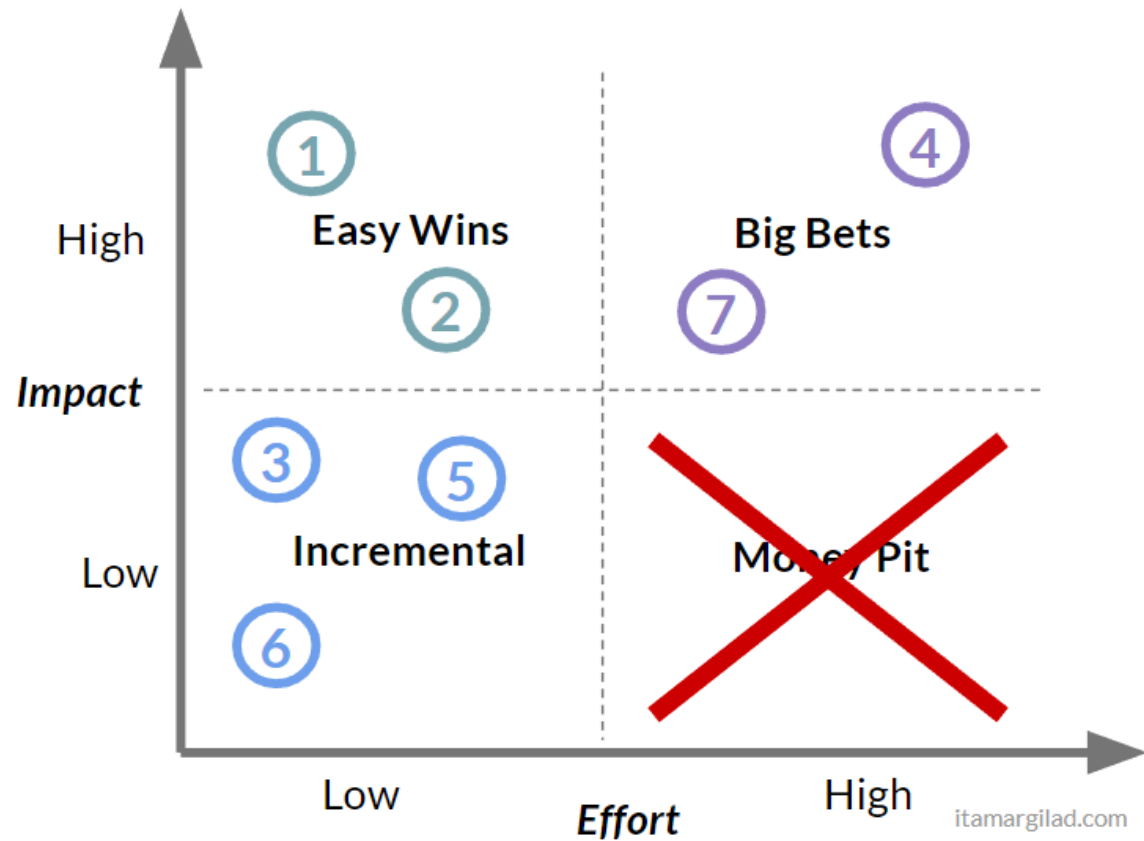
Open Multi-voting



Secret Ballot Voting



Benefit/Effort Diagram



Pros/Cons Analysis

Pros and Cons List Template

<https://www.vertex42.com/ExcelTemplates/pros-and-cons-list.html>



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PROS		56	49	CONS	
List	Value (0-10)		Value (0-10)	List	
Location	3			4	Distance from Family
New House	5			5	Miss Friends
Schools	8			6	Traffic
Grocery Store Access	2			5	Hassles
Church	7			8	Crime Rate
Mall	4			3	Move Expense
Theater Access	6			2	More Urban
More Activities	3			6	Unknown Problems
Better Job	7			3	Living Expense Higher
Change of Scene	2			2	Stress
Beach Access	4			5	Change Schools
Change Schools	5				

Other Evaluation Techniques

- Cognitive Walkthrough
- Heuristic Evaluation
- Review-based Evaluation
- Lab studies involving users
- Field studies involving users

Cognitive Walkthrough

- evaluates design on how well it supports user in learning task
- usually performed by expert in cognitive psychology
- expert ‘walks through’ design to identify potential problems using psychological principles
- forms used to guide analysis

Cognitive Walkthrough (ctd)

- For each task walkthrough considers
 - what impact will interaction have on user?
 - what cognitive processes are required?
 - what learning problems may occur?
- Analysis focuses on goals and knowledge: does the design lead the user to generate the correct goals?

Heuristic Evaluation

- usability criteria (heuristics) are identified
 - UX laws
 - UI principles and patterns
- design examined by experts to see if these are violated
- Example heuristics
 - system behaviour is predictable
 - system behaviour is consistent
 - feedback is provided
- Heuristic evaluation `debugs' design.

Review-based evaluation

- Results from the literature used to support or refute parts of design.
- Care needed to ensure results are transferable to new design.
- Model-based evaluation
- Cognitive models used to filter design options
- Design rationale can also provide useful evaluation information

Laboratory studies

- Advantages:
 - specialist equipment available
 - uninterrupted environment
- Disadvantages:
 - lack of context
 - difficult to observe several users cooperating
- Appropriate
 - if system location is dangerous or impractical for constrained single user systems to allow controlled manipulation of use

Field Studies

- Advantages:
 - natural environment
 - context retained (though observation may alter it)
 - longitudinal studies possible
- Disadvantages:
 - distractions
 - noise
- Appropriate
 - where context is crucial for longitudinal studies

Case Study: Concept Selection

Reusable Insulin Syringe



Concept Selection Criteria

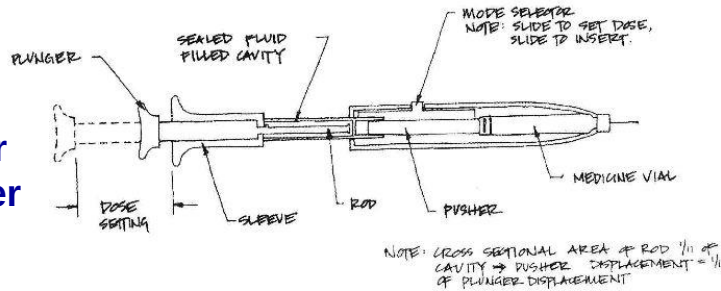
- Ease of handling
- Ease of use
- Readability of dose settings
- Dose metering accuracy
- Durability
- Ease of manufacture
- Portability

Concept Selection Process

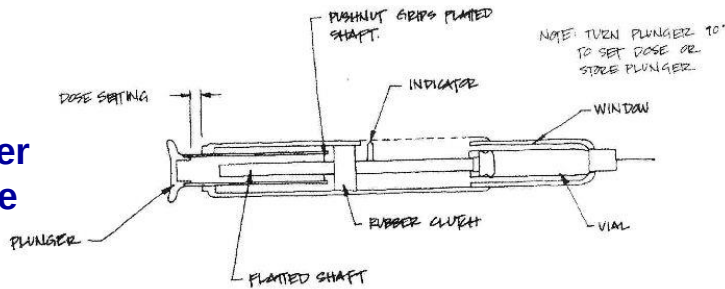
- Prepare the Matrix
 - Criteria
 - Reference Concept
 - Weightings
- Rate Concepts
 - Scale (+ – 0) or (1–5)
 - Compare to Reference Concept or Values
- Rank Concepts
 - Sum Weighted Scores
- Combine and Improve
 - Remove Bad Features
 - Combine Good Qualities
- Select Best Concept
 - May Be More than One
 - Beware of Average Concepts
- Reflect on the Process
 - Continuous Improvement

Syringe Example: Concepts

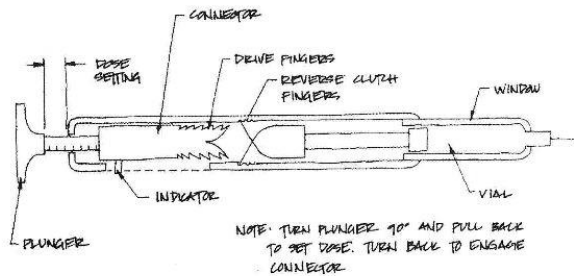
**A.
Master
Cylinder**



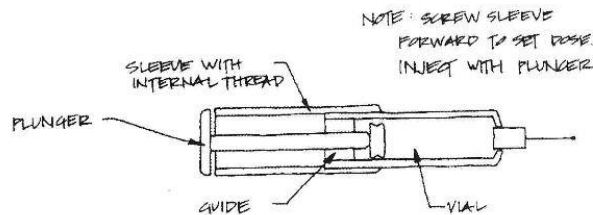
**B.
Rubber
Brake**



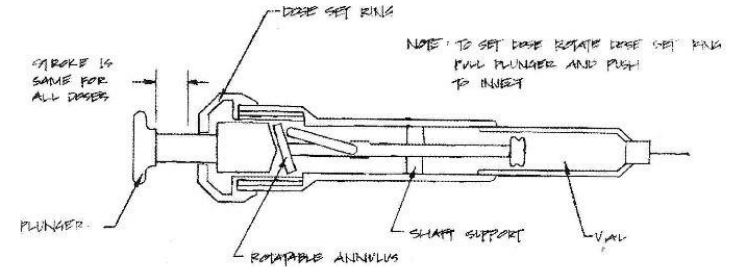
**C.
Ratchet**



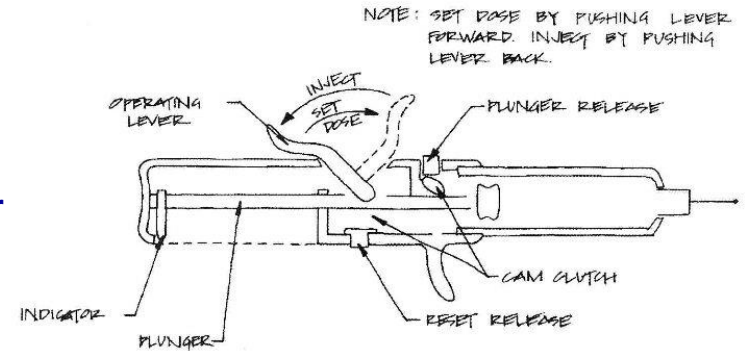
**D.
Plunge
Stop**



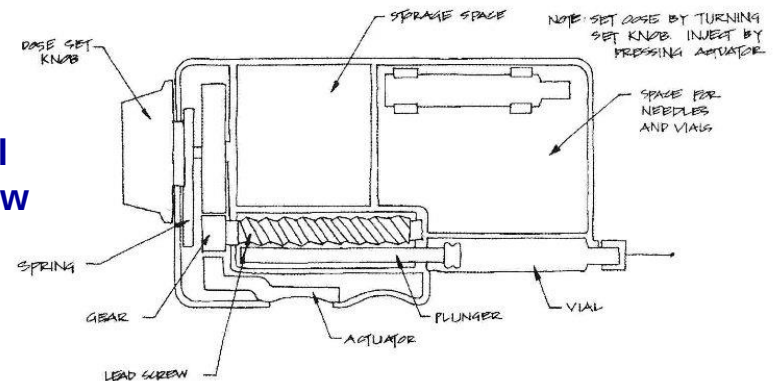
**E.
Swash
Ring**



**F.
Lever
Set**



**G.
Dial
Screw**



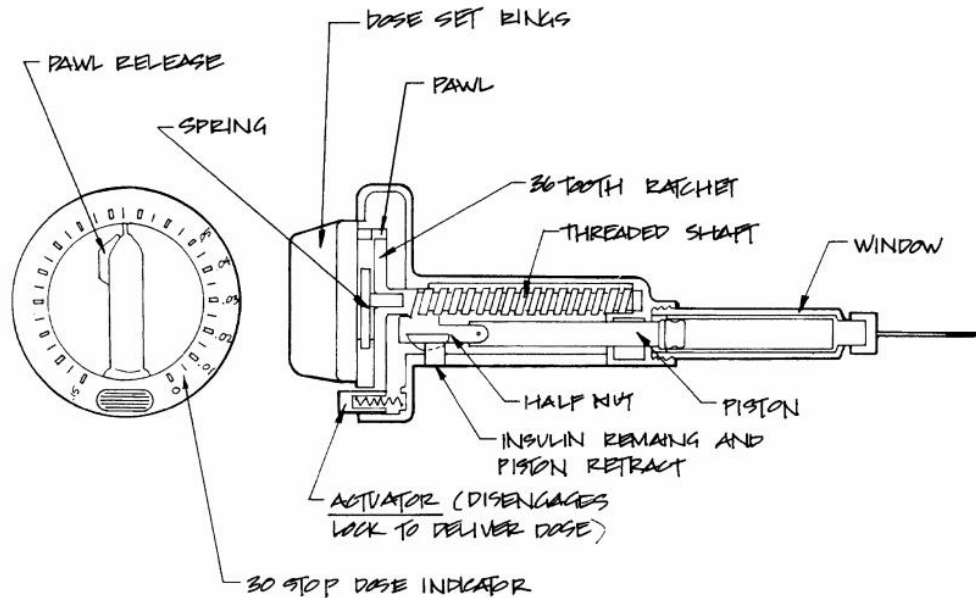
Syringe Example: Concept Screening Matrix

	Concepts							
	A	B	C	D	E	F	G	REF.
Selection Criteria	Master Cylinder	Rubber Brake	Ratchet	Plunge Stop	Swash Ring	Lever Set	Dial Screw	Existing Product
Ease of Handling	0	0	–	0	0	–	–	0
Ease of Use	0	–	–	0	0	+	0	0
Readability of Settings	0	0	+	0	+	0	+	0
Dose Metering Accuracy	+	+	+	+	+	0	+	0
Durability	0	0	0	0	0	+	0	0
Ease of Manufacture	+	–	–	0	0	–	0	0
Portability	+	+	–	–	0	–	–	0
Plusses	3	2	2	1	2	2	2	0
Sames	4	3	1	5	5	2	3	0
Minuses	0	2	4	1	0	3	2	0
Net	3	0	-2	0	2	-1	0	
Rank	1	3	7	5	2	6	4	
Continue?	Yes	Yes	No	No	Yes	No	Yes	

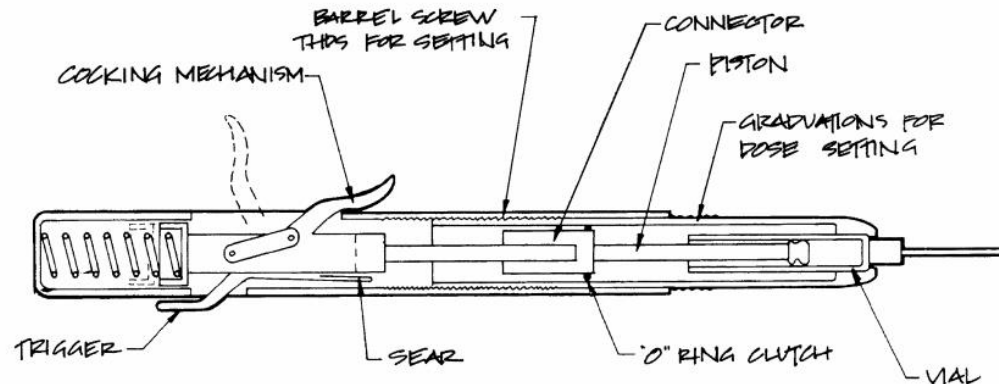
Syringe Example:

Combine and Improve the Concepts

Concept G+:
Dial Screw+



Concept DF:
Lever Stop



Syringe Example: Concept Scoring Matrix

		Concepts							
		A (reference) Master Cylinder		DF Lever Stop		E Swash Ring		G+ Dial Screw+	
Selection Criteria	Weight	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score	Rating	Weighted Score
Ease of Handling	5%	3	0.15	3	0.15	4	0.2	4	0.2
Ease of Use	15%	3	0.45	4	0.6	4	0.6	3	0.45
Readability of Settings	10%	2	0.2	3	0.3	5	0.5	5	0.5
Dose Metering Accuracy	25%	3	0.75	3	0.75	2	0.5	3	0.75
Durability	15%	2	0.3	5	0.75	4	0.6	3	0.45
Ease of Manufacture	20%	3	0.6	3	0.6	2	0.4	2	0.4
Portability	10%	3	0.3	3	0.3	3	0.3	3	0.3
Total Score		2.75		3.45		3.10		3.05	
Rank		4		1		2		3	
Continue?		No		Develop		No		No	

Concept Selection Caveats

- Decomposition of concept quality
 - Determining independent selection criteria which may indicate the concept quality is not straightforward, but significantly impact the process.
- Beware of subjective criteria
 - Criteria such as aesthetics, ease of use, or even ergonomics
- Include production cost into your selection criteria

Concept Selection Exercise: Mechanical Pens



Concept Testing

Concept Testing is Used for Several Purposes

- Go/no-go decisions
- What market to be in?
- Selecting among alternative concepts
- Confirming concept selection decision
- Benchmarking
- Soliciting improvement ideas
- Forecasting demand
- Ready to launch?

Concept Testing Process

1. Define the purpose of the test
2. Choose a survey population
3. Choose a survey format
4. Communicate the concept
5. Measure customer response
6. Interpret the results
7. Reflect on the results and the process

Concept Testing Example: emPower Electric Scooter



Scooter Example

1. Purpose of concept test:

- What market to be in?

2. Sample populations:

- College students who live 1-3 miles from campus
- Factory transportation

3. Survey format:

- Face-to-face interviews

Communicating the Concept

- Verbal description
- Sketch
- Photograph or rendering
- Storyboard
- Video
- Simulation
- Interactive multimedia
- Physical appearance model
- Working prototype

least useful

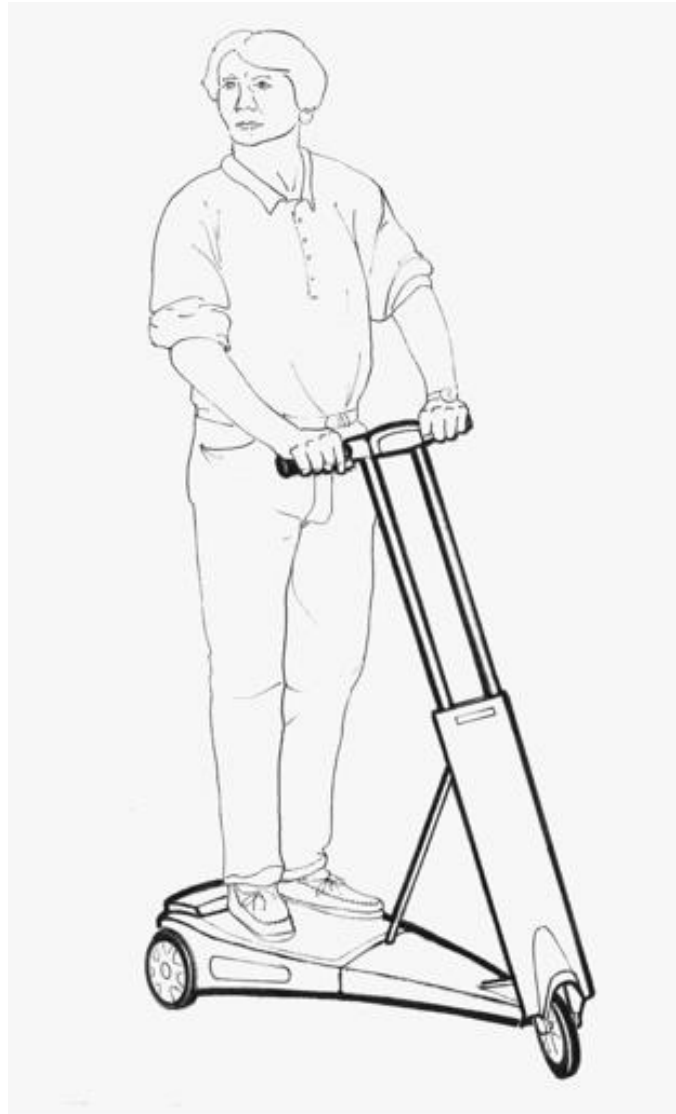


most useful

Verbal Description

- The product is a lightweight electric scooter that can be easily folded and taken with you inside a building or on public transportation.
- The scooter weighs about 25 pounds. It travels at speeds of up to 15 miles per hour and can go about 12 miles on a single charge.
- The scooter can be recharged in about two hours from a standard electric outlet.
- The scooter is easy to ride and has simple controls — just an accelerator button and a brake.

Sketch



Rendering



Storyboard

EMpower *transPort*

compact storage

portable
transportation

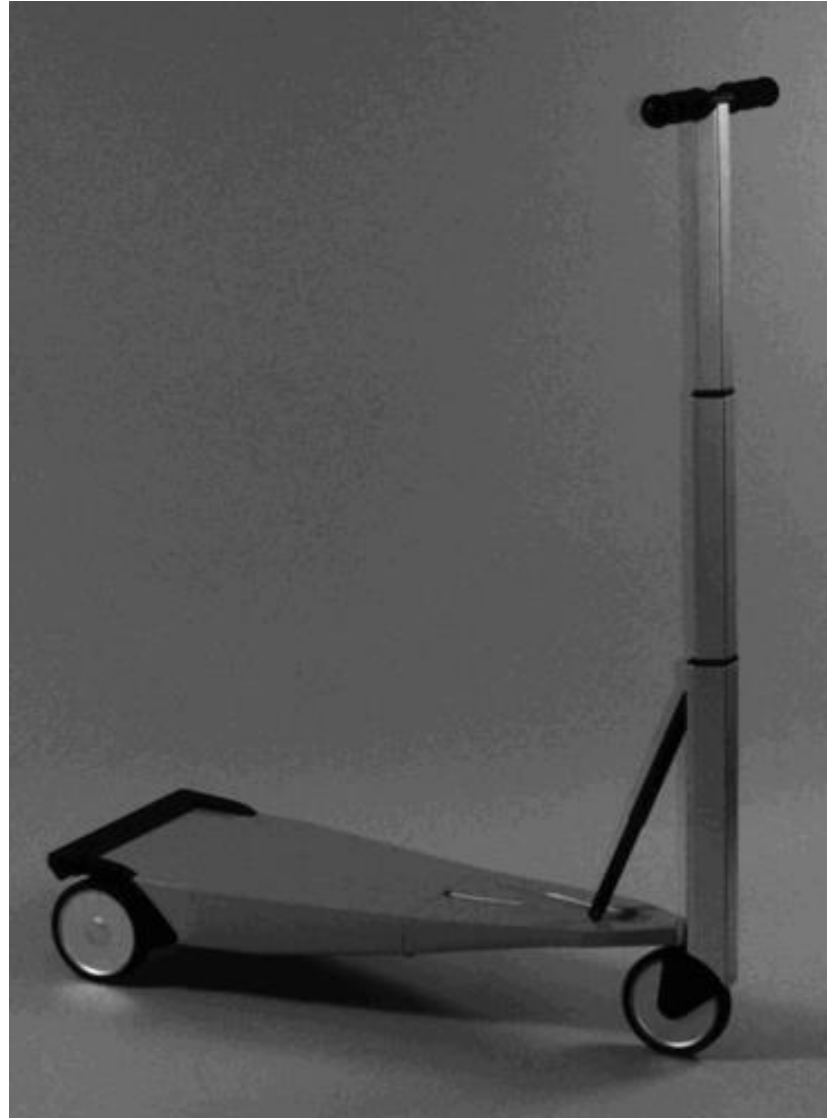
reduced
transit
time



3D Solid CAD Model



Appearance Model



Working (Alpha) Prototype



Pre-Production (Beta) Prototype



Video, Animation, Interactive Multimedia, or Live Demonstration



Survey Format

- **PART 1, Qualification**
 - How far do you live from campus?
 - <If not 1-3 miles, thank the customer and end interview.>
 - How do you currently get to campus from home?
 - How do you currently get around campus?
- **PART 2, Product Description**
 - <Present the concept description.>

Survey Format

- **PART 3, Purchase Intent**

- If the product were priced according to your expectations, how likely would you be to purchase the scooter within the next year?

☐

I would
definitely not
purchase
the scooter.

☐

I would
probably not
purchase
the scooter.

☐

I **might**
or might not
purchase
the scooter.

☐

I would
probably
purchase
the scooter.

☐

I would
definitely
purchase
the scooter.

↑
“second box”

↑
“top box”

Survey Format

- **PART 4, Comments**
 - What would you expect the price of the scooter to be?
 - What concerns do you have about the product concept?
 - Can you make any suggestions for improving the product concept?
- **Thank you.**

Interpreting the Results: Forecasting Sales

$$Q = N \times A \times P$$

- Q = sales (annual)
- N = number of (annual) purchases
- A = awareness x availability (fractions)
- P = probability of purchase (surveyed)

$$= C_{\text{def}} \times F_{\text{def}} + C_{\text{prob}} \times F_{\text{prob}}$$

↑
“top box”

↑
“second box”

C_{def} and C_{prob} – calibration coefficients, $C_{\text{def}} = [0.1, 0.5]$, $C_{\text{prob}} = [0, 0.3]$,
 F_{def} and F_{prob} – fraction who answer “definitely” and “probably”

Forecasting Example: College Student Market

- N = off-campus grad students (200,000)
- A = 0.2 (realistic) to 0.8 (every bike shop)
- $P = 0.4 \times \textit{top-box} + 0.2 \times \textit{second-box}$
- $Q =$
- Price point \$795

Forecasting Example: Factory Transport Market

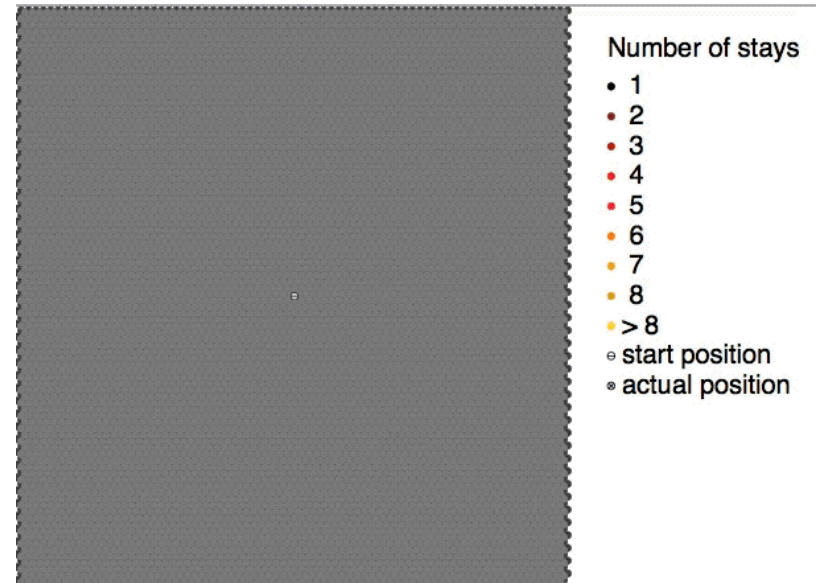
- N = current bicycle and scooter sales to factories (150,000)
- $A = 0.25$ (single distributor's share)
- $P = 0.4 \times \textit{top-box} + 0.2 \times \textit{second-box}$
- $Q = 150,000 \times 0.25 \times [0.4 \times 0.3 + 0.2 \times 0.2]$
= 6000 units/yr
- Price point \$1500

Pricing Options

Engineering Idea #4

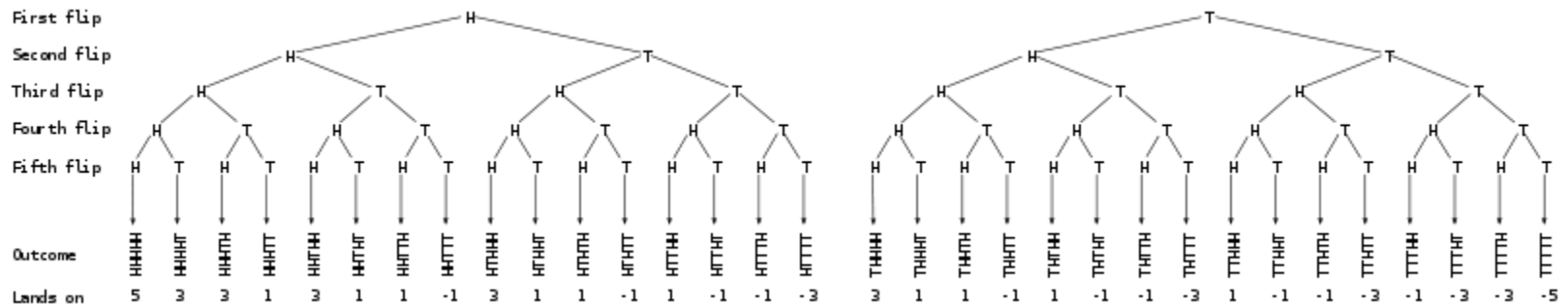
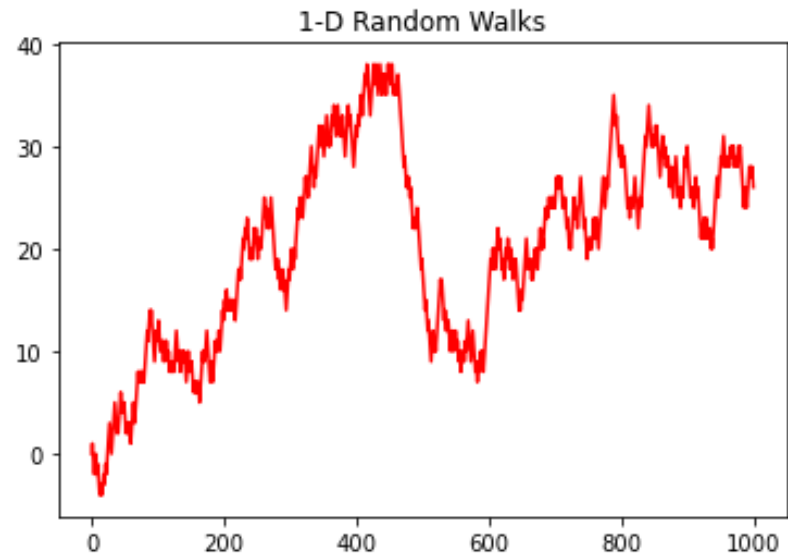
Brownian Motion

- Random motion of particles named after Robert Brown who first described the phenomenon in 1827 while looking through a microscope at pollen of the plant immersed in water.
- Mathematically described as a Wiener process, W_t :
 1. $W_0 = 0$
 2. W_t is almost certainly continuous
 3. W_t has independent increments
 4. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s \leq t$



Random Walk

1. Initialize $t = 0, S_t = 0$
2. Repeat n times
 - Throw a fair coin:
 - If head, $S_{t+1} = S_t + 1$
 - Otherwise, $S_{t+1} = S_t - 1$

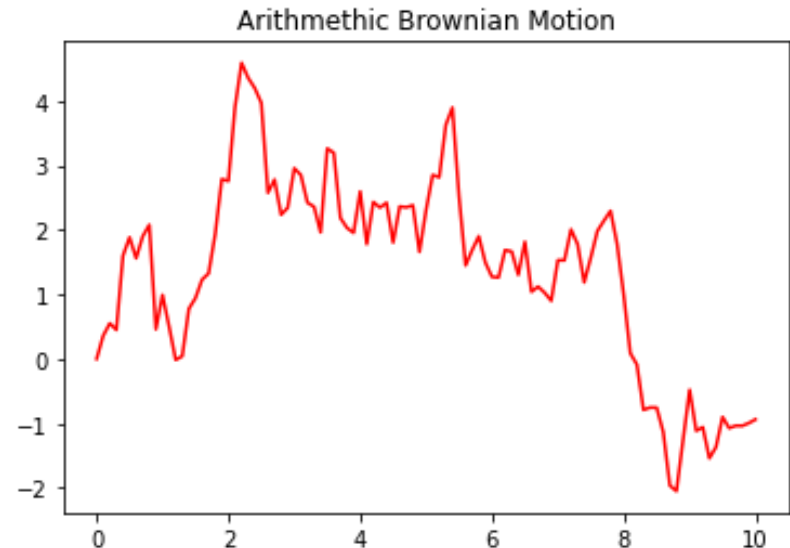


Arithmetic Brownian Motion

1. Initialize $t = 0, S_t = 100, \Delta t = 0.1$
2. Repeat n times
 1. Draw $\epsilon \sim N(0,1)$
 2. $S_{t+\Delta t} = S_t + \sigma\epsilon\sqrt{\Delta t}$

Where σ calculated from historical prices:

$$\sigma = \frac{1}{\sqrt{n}} \sum_{i=0}^m (\bar{S} - S_i)^2$$

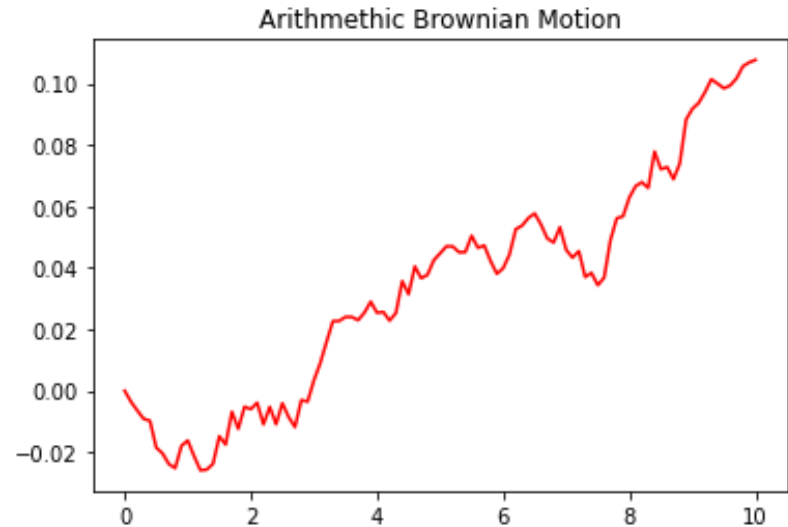


Time Value of Money

- Money **today** worth **more than tomorrow**, because of number factors:
 - constant inflation
 - risk-free rate
 - other investment opportunities
- For one year
 - $S_1 = S_0(1 + r)$
 - r – risk-free investment rate (per year)
- For n years compounded annually:
 - $S_n = S_0(1 + r)^n$
- For n years compounded monthly
 - $S_n = S_0(1 + r/12)^{n/12}$
- For n years compounded continuously
 - $S_n = S_0e^{rn}$

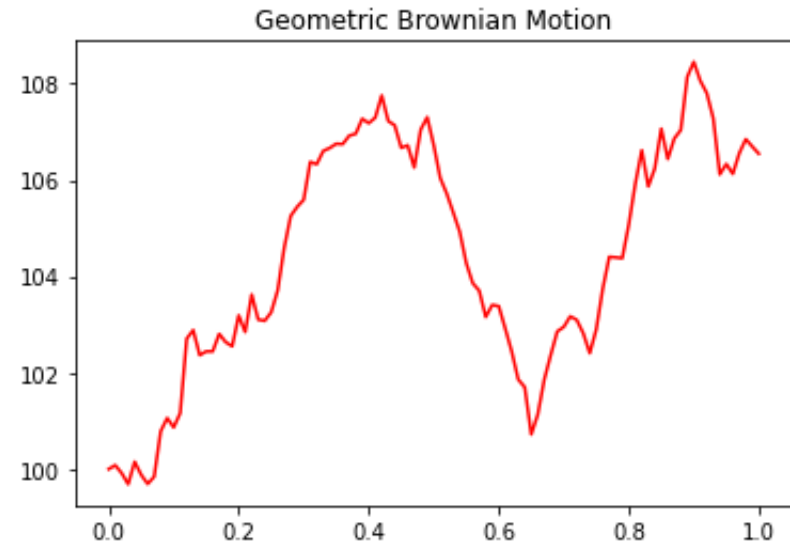
Arithmetic Brownian Motion with Drift

1. Initialize $t = 0, S_t = 100, \Delta t = 0.1$
2. Repeat n times
 1. Draw $\epsilon \sim N(0,1)$
 2. $S_{t+\Delta t} = S_t + S_t r \Delta t + \sigma \epsilon \sqrt{\Delta t}$



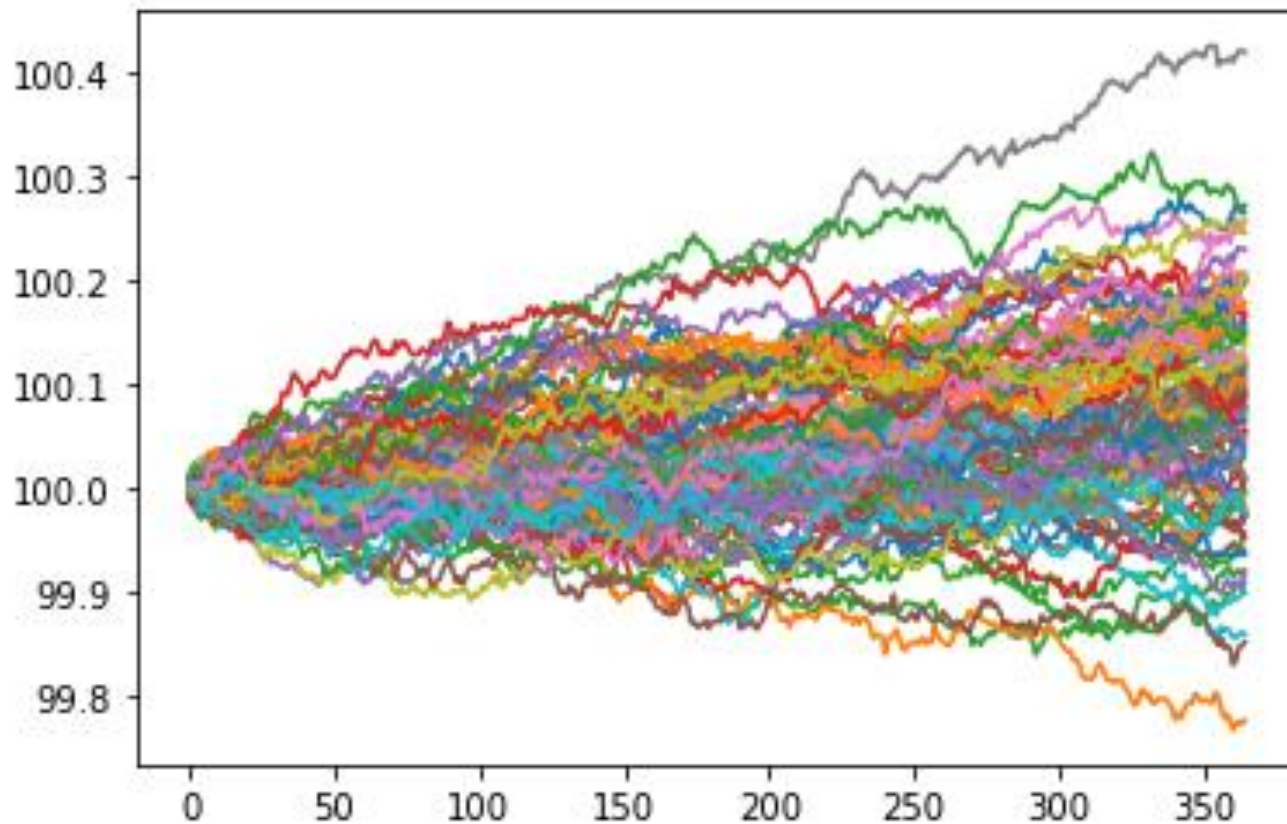
Geometric Brownian Motion with Drift

1. Initialize $t = 0, S_t = 100, \Delta t = 0.1$
2. Repeat n times
 1. Draw $\epsilon \sim N(0,1)$
 2. $S_{t+\Delta t} = S_t \exp(r\Delta t + \sigma\epsilon\sqrt{\Delta t})$



Modelling Asset Prices as Geometric Brownian Motion

- Monte-Carlo Simulation



Options

Options are financial contracts that give right, but not an obligation, to buy (**call option**) or sell (**put option**) certain asset in future on agreed price (**strike price**) in exchange for some premium.



I produce corn, and it is important that the next year corn price will be above my production costs.

I need put option.

I see Apple is going to release new product next year and its stock should rise. Buying now is risky, but I want today's price when it grew.

I need call option.



Option Pricing

- Option price is usually derived from several parameters such as the **performance of its underlying asset** S_t , **risk-free rate** r and **strike price** K .
- European options can be exercised only on their maturity date T .
- Option's intrinsic value is bound within a range depending on K and S_t

$$\max(S_t - K, 0) \leq c \leq S_t$$

$$\max(K - S_t, 0) \leq p \leq K$$

Option Pricing Algorithm

1. Set current asset price as S_0
2. Find series of $\log(S_i)$ for the historic period
3. Calculate standard deviation σ for log-returns $\log(S_i)$
4. Set r as risk-free rate
5. Simulate many instances of GBM for period T
6. Calculate discounted option prices for each k^{th} price path using intrinsic value of option at time T
 1. Call option: $\max(K - S_T) * e^{-rT}$
 2. Put option: $\max(S_T - K) * e^{-rT}$
7. Average k values and return it as risk-neutral value of option

Further Reading

- Hull, J. C. (2003). *Options futures and other derivatives*. Pearson Education India.
- Gilli, M., Maringer, D., & Schumann, E. (2019). *Numerical methods and optimization in finance*. Academic Press.